

Software Testing Strategies

1 Organizing for Software Testing

Software testing is the process of executing a program with the intent of finding errors.

For every software project, an inherent **conflict of interest** exists when developers test their own software.

1.1 Psychological Aspect of Testing

- Software development (analysis, design, coding) is a **constructive activity**.
- Testing is a **destructive activity** because it attempts to break what was built.
- Developers subconsciously design tests to **prove correctness**, not to find defects.

If developers do not find the errors, the customer eventually will.

1.2 Common Misconceptions

The following assumptions are **incorrect**:

- Developers should not test their own software.
- Software should be thrown to outsiders for testing only.
- Testers should get involved only after coding is complete.

2 Role of Developers and Independent Test Group (ITG)

2.1 Developer Responsibilities

- Perform **unit testing** on individual components.
- Often perform **integration testing**.
- Fix defects found during testing.

2.2 Independent Test Group (ITG)

An **Independent Test Group (ITG)** is a team responsible for unbiased testing of the software.

- Removes conflict of interest.
- Paid specifically to find defects.
- Often reports to Software Quality Assurance (SQA).

Developers and ITG work closely throughout the project; testing is not a handoff activity.

3 Software Testing Strategy: The Big Picture

Software testing can be viewed using the **spiral model**.

3.1 Development View (Inward Spiral)

- System Engineering
- Requirements Analysis
- Design
- Coding

Each turn inward reduces the level of abstraction.

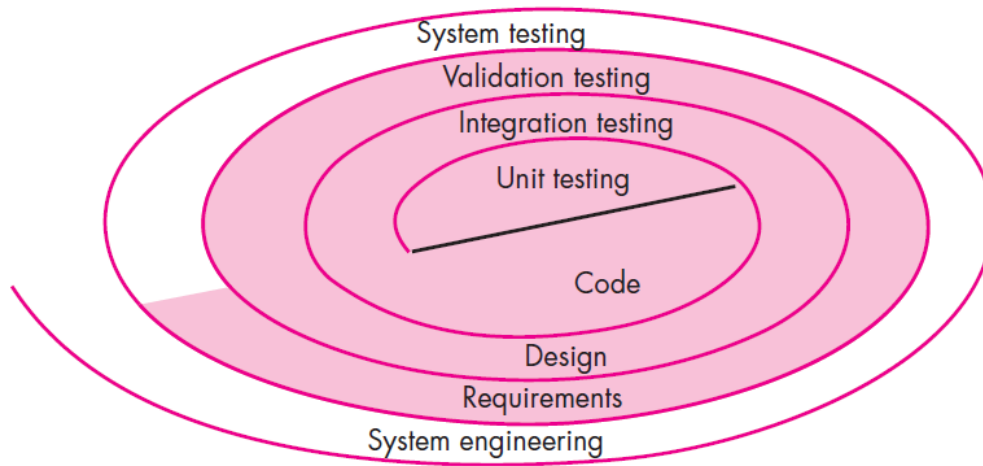
3.2 Testing View (Outward Spiral)

- Unit Testing
- Integration Testing
- Validation Testing
- System Testing

Each turn outward increases the scope of testing.

Development proceeds inward; testing proceeds outward.

4 Software Testing Levels



4.1 Unit Testing

Unit testing focuses on individual components, classes, or functions.

Key Characteristics:

- Performed by developers
- Focuses on control paths and logic
- Ensures each unit works correctly

4.2 Integration Testing

Integration testing focuses on interactions between integrated components.

- Ensures components work together
- Focus on inputs and outputs
- Identifies interface defects

4.3 Validation Testing

Validation testing ensures the software meets customer requirements.

- Based on requirements specification
- Answers: *“Are we building the right product?”*

4.4 System Testing

System testing verifies the complete system in its operational environment.

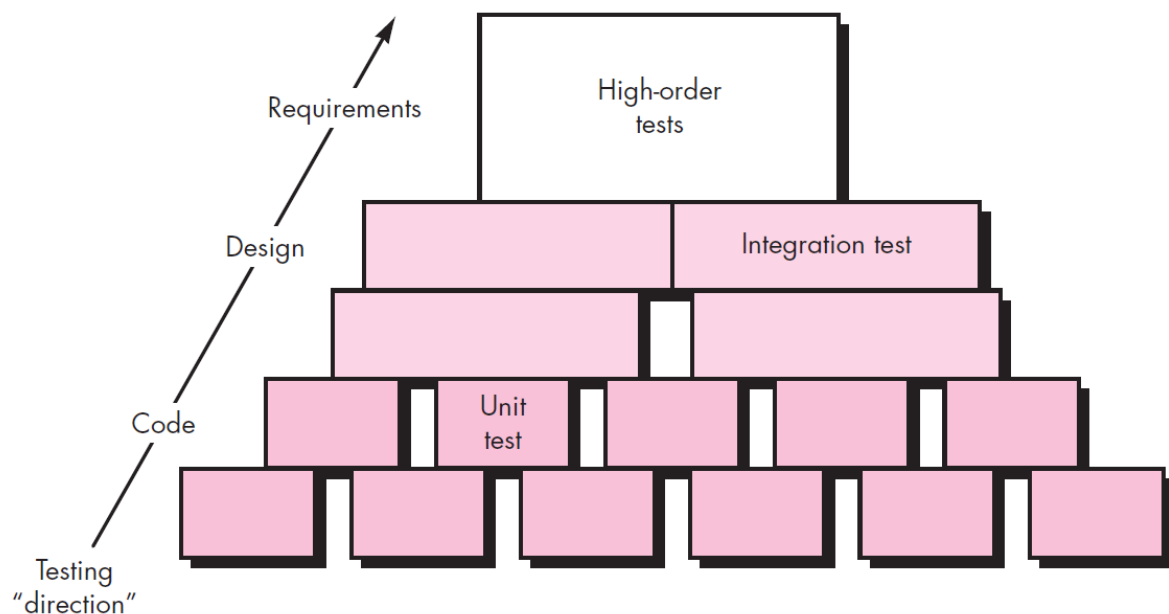
- Includes hardware, people, databases
- Tests overall system performance and behavior

Level	Focus	Performed By
Unit Testing	Individual components	Developers
Integration Testing	Component interaction	Developers / ITG
Validation Testing	Requirements	ITG / Users
System Testing	Whole system	ITG

5 Procedural View of Testing

From a procedural perspective, testing occurs in four sequential steps:

1. Unit Testing
2. Integration Testing
3. Validation Testing
4. System Testing



This aligns with the layered testing diagram where testing moves upward from code to requirements.

6 Criteria for Completion of Testing

A fundamental question: “**When are we done testing?**”

6.1 Pragmatic Answers

- Testing never truly ends; users continue testing during use.
- Testing stops when time or budget runs out.

6.2 More Rigorous Approaches

- Cleanroom Software Engineering
- Statistical usage testing
- Software reliability models

Metrics and reliability models provide better guidance than intuition alone.

7 Key Quotes and Takeaways

- “Testing is the unavoidable part of any responsible effort to develop software.”
- “Optimism is the occupational hazard of programming; testing is the treatment.”
- Testing does not ensure quality—**engineering does**.

8 Summary

- Testing is a structured, multi-level process.
- Developers and independent testers play complementary roles.
- Testing mirrors development in reverse order.
- Completion of testing requires practical and statistical judgment.